

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in physicochemical characterization, stability analysis, and DOE-guided process development (FT-IR, UV-Vis-NIR, NMR/GPC, TGA/DSC, QCM-D, electrochemistry). Currently leading bio-based polymer scale-up and translating structure-property data into reproducible process windows.

CORE COMPETENCIES

Physicochemical Characterization

Pre-Formulation Materials Science

Thermal & Spectroscopic Analysis

Polymer/Solid-State Systems

Stability Analysis

DOE/SPC Process Development

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, defining reproducible process windows and scale-up specifications from material-quality data.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and characterized organic electrochromic materials, mapping structure-property-stability relationships across 15+ variants using spectroscopy, electrochemistry, thermal analysis, NMR, and GPC.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded molecular thin-film coating process development, tuning CVD and surface preparation to control morphology, interfaces, and optical performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE and in-situ QCM-D monitoring to define growth kinetics and reproducible process controls.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochromic Material Reliability Study Chemistry of Materials 2025

Diagnosed degradation and instability in organic materials under accelerated stress conditions.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC

AFOSR Defense Thin-Film Coatings DoD Funded

Connected CVD and surface preparation parameters with morphology, interface quality, and device performance.

CVD, surface preparation, thin films

Bio-Based Polymer Scale-Up Process Current

Defined DOE-guided process windows and material-quality specifications for reproducible polymer scale-up.

DOE, SPC, polymer characterization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Characterization: FT-IR, UV-Vis-NIR, NMR, GPC/SEC, TGA, DSC, QCM-D, cyclic voltammetry, spectroelectrochemistry, GC-MS, LC-MS

Materials & Stability: Organic/small-molecule materials, polymers, thin films, CVD, surface functionalization, degradation/failure analysis

Process & Data: DOE, SPC, JMP (basic), Python (basic), Excel, process optimization, reproducibility, technical documentation

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)